

Table 1: LSDA improvement over ablations for configuration dependency *identification*.

System	w/o V	w/o C	w/o M	w/o Sym	w/o BNF
	Mean (impr.)	Mean (impr.)	Mean (impr.)	Mean (impr.)	Mean (impr.)
Precision					
HDFS	0.072 (344.6%)	0.409 (-13.7%)	0.333 (40.8%)	0.044 (829.6%)	0.370 (1.9%)
HBASE	0.062 (584.0%)	0.400 (6.0%)	0.500 (-15.2%)	0.029 (1362.1%)	0.378 (12.2%)
YARN	0.270 (101.5%)	0.608 (-10.5%)	0.549 (-0.9%)	0.086 (532.6%)	0.628 (-13.4%)
MAPREDUCE	0.092 (323.9%)	0.614 (-36.6%)	0.274 (42.0%)	0.068 (471.0%)	0.380 (2.4%)
CORE	0.051 (960.8%)	0.625 (-13.4%)	0.289 (87.2%)	0.031 (1645.2%)	0.532 (1.7%)
GLANCE	0.358 (32.5%)	0.441 (7.5%)	0.203 (133.5%)	0.085 (457.6%)	0.425 (11.5%)
CINDER	0.269 (35.0%)	0.546 (-33.5%)	0.268 (35.4%)	0.050 (626.0%)	0.346 (4.9%)
Average	0.167 (344.6%)	0.463 (-13.7%)	0.345 (40.8%)	0.056 (829.6%)	0.423 (1.9%)
Recall					
HDFS	0.600 (10.7%)	0.215 (208.8%)	0.539 (23.2%)	0.412 (61.2%)	0.406 (63.6%)
HBASE	0.485 (78.9%)	0.724 (19.9%)	0.098 (785.7%)	0.469 (85.1%)	0.773 (12.3%)
YARN	0.683 (37.6%)	0.539 (74.5%)	0.595 (58.0%)	0.358 (162.6%)	0.674 (39.5%)
MAPREDUCE	0.468 (90.0%)	0.554 (60.6%)	0.464 (91.8%)	0.239 (272.4%)	0.749 (18.8%)
CORE	0.060 (1263.3%)	0.302 (171.2%)	0.048 (1604.2%)	0.031 (2538.7%)	0.729 (12.2%)
GLANCE	0.368 (148.0%)	0.741 (23.2%)	0.381 (139.5%)	0.070 (1204.3%)	0.877 (4.1%)
CINDER	0.213 (317.2%)	0.766 (16.1%)	0.198 (349.0%)	0.071 (1152.1%)	0.886 (0.3%)
Average	0.343 (277.8%)	0.549 (82.1%)	0.332 (435.4%)	0.235 (781.4%)	0.728 (21.6%)
F1-score					
HDFS	0.128 (256.3%)	0.280 (62.9%)	0.410 (11.2%)	0.079 (477.2%)	0.385 (18.4%)
HBASE	0.109 (383.5%)	0.507 (3.9%)	0.152 (246.7%)	0.054 (875.9%)	0.504 (4.6%)
YARN	0.386 (78.2%)	0.568 (21.1%)	0.570 (20.7%)	0.138 (398.6%)	0.643 (7.0%)
MAPREDUCE	0.153 (252.9%)	0.574 (-6.1%)	0.341 (58.1%)	0.106 (408.5%)	0.500 (7.8%)
CORE	0.055 (1074.5%)	0.407 (58.7%)	0.078 (728.2%)	0.031 (1983.9%)	0.611 (5.7%)
GLANCE	0.357 (73.9%)	0.538 (15.4%)	0.478 (29.9%)	0.075 (728.0%)	0.571 (8.8%)
CINDER	0.232 (121.0%)	0.632 (-18.8%)	0.224 (129.0%)	0.058 (784.5%)	0.496 (3.4%)
Average	0.203 (318.7%)	0.501 (19.6%)	0.319 (175.4%)	0.077 (807.6%)	0.530 (8.0%)
AUC					
HDFS	0.797 (4.4%)	0.607 (37.1%)	0.769 (8.2%)	0.703 (18.4%)	0.703 (18.4%)
HBASE	0.739 (26.3%)	0.861 (8.4%)	0.549 (69.9%)	0.727 (28.3%)	0.886 (5.3%)
YARN	0.840 (15.5%)	0.769 (26.2%)	0.797 (21.7%)	0.676 (43.5%)	0.837 (15.9%)
MAPREDUCE	0.729 (29.4%)	0.777 (21.5%)	0.731 (29.1%)	0.616 (53.2%)	0.873 (8.1%)
CORE	0.529 (71.9%)	0.651 (39.6%)	0.524 (73.4%)	0.514 (76.8%)	0.864 (5.2%)
GLANCE	0.682 (40.1%)	0.868 (10.1%)	0.690 (38.5%)	0.532 (79.7%)	0.935 (2.3%)
CINDER	0.606 (55.8%)	0.882 (7.0%)	0.599 (57.6%)	0.534 (76.8%)	0.942 (0.2%)
Average	0.703 (34.8%)	0.772 (21.4%)	0.666 (42.7%)	0.615 (53.7%)	0.877 (7.9%)

Table 2: LSDA improvement over ablations for configuration dependency *classification*.

System	w/o V	w/o C	w/o M	w/o Sym	w/o BNF
	Mean (impr.)	Mean (impr.)	Mean (impr.)	Mean (impr.)	Mean (impr.)
Precision					
HDFS	0.041 (9584.6%)	0.256 (-15.2%)	0.162 (68.6%)	0.055 (318.2%)	0.230 (0.0%)
HBASE	0.000 ()	0.222 (-12.2%)	0.200 (-2.5%)	0.055 (254.5%)	0.232 (-15.9%)
YARN	0.145 (180.0%)	0.385 (5.5%)	0.308 (31.8%)	0.167 (143.1%)	0.246 (65.0%)
MAPREDUCE	0.031 (751.6%)	0.420 (-37.1%)	0.258 (2.3%)	0.069 (282.6%)	0.315 (-16.2%)
CORE	0.033 (763.6%)	0.389 (-26.7%)	0.115 (147.8%)	0.089 (220.2%)	0.235 (21.3%)
CINDER	0.141 (143.3%)	0.295 (16.3%)	0.086 (298.8%)	0.061 (462.3%)	0.223 (53.8%)
GLANCE	0.179 (-14.0%)	0.263 (-41.4%)	0.268 (-42.5%)	0.112 (37.5%)	0.226 (-31.9%)
Average	0.096 (9584.6%)	0.321 (-15.2%)	0.228 (68.6%)	0.087 (245.9%)	0.244 (10.9%)
Recall					
HDFS	0.264 (85.6%)	0.203 (141.4%)	0.283 (73.1%)	0.201 (143.8%)	0.364 (34.6%)
HBASE	0.020 (1375.0%)	0.409 (-27.9%)	0.029 (917.2%)	0.161 (83.2%)	0.260 (13.5%)
YARN	0.379 (58.1%)	0.322 (86.0%)	0.338 (77.2%)	0.115 (420.9%)	0.296 (102.4%)
MAPREDUCE	0.225 (70.2%)	0.322 (19.0%)	0.269 (42.4%)	0.067 (472.3%)	0.445 (-13.9%)
CORE	0.100 (379.0%)	0.190 (152.1%)	0.051 (839.2%)	0.033 (1351.5%)	0.269 (78.1%)
CINDER	0.075 (569.3%)	0.398 (26.1%)	0.086 (483.7%)	0.101 (397.0%)	0.520 (-3.5%)
GLANCE	0.117 (245.3%)	0.272 (48.5%)	0.155 (160.6%)	0.135 (199.6%)	0.336 (20.2%)
Average	0.183 (397.5%)	0.331 (63.5%)	0.159 (369.4%)	0.116 (436.1%)	0.356 (33.1%)
F1-score					
HDFS	0.129 (178.3%)	0.183 (96.2%)	0.301 (19.3%)	0.028 (1182.1%)	0.264 (36.0%)
HBASE	0.000 ()	0.472 (-5.5%)	0.053 (741.5%)	0.031 (1338.7%)	0.373 (19.6%)
YARN	0.236 (116.1%)	0.397 (28.5%)	0.340 (50.0%)	0.070 (628.6%)	0.317 (60.9%)
MAPREDUCE	0.083 (352.4%)	0.509 (-26.1%)	0.177 (112.4%)	0.029 (1196.6%)	0.379 (-0.8%)
CORE	0.034 (1452.9%)	0.347 (52.2%)	0.037 (1327.0%)	0.015 (3420.0%)	0.375 (40.8%)
CINDER	0.187 (203.7%)	0.495 (14.7%)	0.183 (210.3%)	0.014 (3957.1%)	0.483 (17.6%)
GLANCE	0.197 (49.7%)	0.279 (5.7%)	0.181 (63.0%)	0.032 (821.9%)	0.376 (-21.5%)
Average	0.124 (394.8%)	0.326 (23.6%)	0.173 (361.4%)	0.031 (1804.7%)	0.367 (21.7%)
AUC					
HDFS	0.713 (6.0%)	0.572 (32.2%)	0.696 (8.6%)	0.543 (39.2%)	0.627 (20.6%)
HBASE	0.500 (63.6%)	0.851 (-3.9%)	0.515 (58.8%)	0.552 (48.1%)	0.711 (15.0%)
YARN	0.713 (19.8%)	0.694 (23.1%)	0.682 (25.2%)	0.558 (53.0%)	0.672 (27.1%)
MAPREDUCE	0.604 (31.3%)	0.736 (7.7%)	0.614 (29.1%)	0.519 (52.8%)	0.742 (6.9%)
CORE	0.516 (62.6%)	0.634 (32.3%)	0.511 (64.3%)	0.505 (66.1%)	0.714 (17.5%)
CINDER	0.587 (58.5%)	0.816 (14.0%)	0.588 (58.1%)	0.506 (83.8%)	0.887 (4.8%)
GLANCE	0.599 (29.7%)	0.701 (10.8%)	0.577 (34.2%)	0.509 (52.0%)	0.792 (-1.9%)
Average	0.605 (38.8%)	0.729 (16.7%)	0.606 (39.8%)	0.527 (56.6%)	0.736 (12.9%)

Table 3: LSDA improvement over ablations for configuration dependency *formalization* correctness.

System	w/o V	w/o C	w/o M
	Mean (improvement)	Mean (improvement)	Mean (improvement)
HDFS	0.250 (88.0%)	0.380 (23.7%)	0.440 (6.8%)
HBASE	0.000 ()	0.590 (6.8%)	0.000 ()
YARN	0.380 (63.2%)	0.550 (12.7%)	0.620 (0.0%)
MAPREDUCE	0.290 (151.7%)	0.650 (12.3%)	0.570 (28.1%)
CORE	0.330 (97.0%)	0.630 (3.2%)	0.500 (30.0%)
CINDER	0.560 (26.8%)	0.730 (-2.7%)	0.670 (6.0%)
GLANCE	0.410 (63.4%)	0.740 (-9.5%)	0.530 (26.4%)
Average	0.317 (101.9%)	0.610 (6.6%)	0.476 (13.6%)